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7. Differentiate the following w.r.t. x :

(i) $\frac{e^x \log x}{x^2}$

(ii) x^{x^2}

(iii) $\tan^{-1} \left(\frac{a-x}{1+ax} \right)$

(iv) $\sqrt{\frac{(x-a)(x-b)}{(x-p)(x-q)}}$

UNIT - IV

8. Evaluate the following:

(i) $\int \frac{x^4 + 1}{x+1} dx$

(ii) $\int \frac{dx}{1+2\cos x}$

9. Evaluate the following:

(i) $\int_0^{\pi/2} x \sin^2 x$

(ii) $\int \sqrt{1+2x-x^2} dx$

Roll No.

97663

BCA 1st Semestwer (New)
Examination - November, 2019

MATHEMATICS

Paper : BCA-103

Time : Three Hours

[Maximum Marks : 80

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt five questions in all, selecting one question from each Unit. Q. No. 1 is compulsory.

1. (a) Given that $A = \{0, 1, 2, 3, 4, 5\}$, $B = \{1, 3, 5\}$. Find $A - B$.

(b) If $\begin{bmatrix} x & 3x-y \\ 2x+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, find x, y, z, w .

(c) If $f(x) = x^5 - \frac{1}{x^5}$, find the value of $f(x) + f\left(\frac{1}{x}\right)$.

(d) Find $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$.

(e) If $y = (1+x)\sqrt{x}$, find $\frac{dy}{dx}$.

(f) If $y = \sin \sqrt{x}$, find $\frac{dy}{dx}$.

(g) Evaluate $\int (3x-4)^{1/3} dx$.

(h) Evaluate $\int \frac{3x^2}{1+x^3} dx$.

UNIT - I

2. (a) Prove that $(A \cup B)' = A' \cap B'$.

(b) In a group of 400 people, 250 can speak English only and 70 can speak Hindi only. Find :

(i) How many can speak both English and Hindi ?

(ii) How many can speak English ?

(iii) How many can speak Hindi ?

3. (a) Prove that :

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$$

(b) Solve $2x + 3y + 3z = 5$, $x - 2y + z = -4$, $3x - y - 2z = 3$.

UNIT - II

4. (a) In $N \times N$, show that the relation defined by $(a, b) R(c, d)$ if and only if $ad = bc$ is an equivalence relation.

(b) If $f, g : R \rightarrow R$ are defined respectively by $f(x) = x^2 + 3x + 1$, $g(x) = 2x - 3$, find (i) fog (ii) gof (iii) fof (iv) gog.

5. (a) Find :

(i) $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{(e^x - 1)}$ (ii) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

(b) Show that $f(x) = \frac{1}{x-a}$ has a discontinuity of second kind at $x = a$.

UNIT - III

6. (a) Derivative of $\sin x$ from first principle.

(b) Differentiate the following w.r.t. x :

(i) $\frac{x^2 - 1}{(x^2 + 7x + 1)}$ (ii) $\frac{2}{x+1} - \frac{x^2}{3x-1}$